SPARTA GLOBAL | AI & TECH INSTITUTE

AI FUNDAMENTALS

UNDERSTANDING MODERN AI: LLMS, RISKS AND CONSIDERATIONS

Investigate the defining models used in modern AI, from the architectures that power them to the strengths, limitations, and reliability challenges of AI systems

LAST SESSION

DEMYSTIFYING AI

Explore how AI systems are built and how they operate, from neural networks and machine learning models to the practical applications and opportunities

TODAY

UNDERSTANDING MODERN AI

Investigate the defining models used in modern AI, from the architectures that power them to the strengths, limitations, and reliability challenges of AI systems

NEXT SESSION

EFFECTIVE AI USE

Develop the skills needed to guide AI effectively through prompt and context engineering, applying critical and responsible judgement throughout

SESSION OVERVIEW

01 **Intro to LLMs**

02 **How LLMs Work**

03 **Performance and Optimisation**

04 **Capability Considerations**

LEARNING OBJECTIVES



- Confidently explain how LLMs work, including the architecture and concepts that underpin their capabilities
- Be able to identify the strengths, limitations, and common failure modes of AI systems, and the approaches used to improve their performance and reliability

01

INTRO TO LLMs

What makes an LLM?

INTRO TO LLMS

WHAT IS IT?

- LLMs is an acronym for Large Language Models
  - They are a type of specialised neural network trained on vast datasets to understand and generate human language

- The LLM name is a direct reference to the massive quantity and type of data they are trained on

- These models learn statistical patterns from the datasets, and use those to generate responses

INTRO TO LLMS

WHAT THEY GENERATE

- LLMs generate text responses, used in:
  - Conversations
  - Fact Finding
  - Code Development
- Programming languages have syntax and rules, and therefore can be generated just as easily as human language

QUESTION

Can you think of any syntax or code rules?

- *Semicolons after each executable line of code (SQL, JavaScript, C)*
- *Import statements at the top of files (Python, Java, C#)*

INTRO TO LLMS

TERMINOLOGY CHECK: LLMs VS LMMs

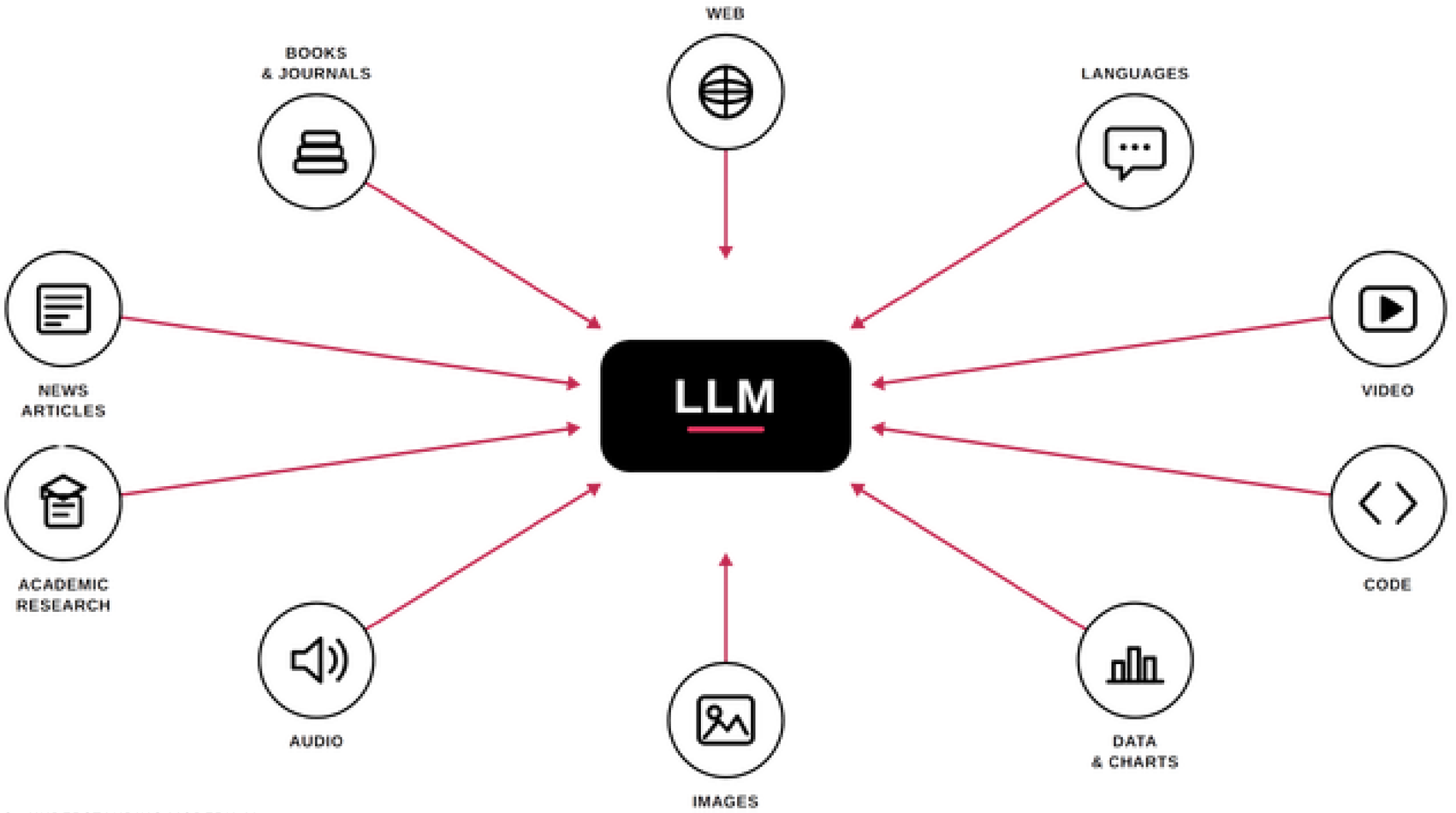
LLMs (Large Language Models) are a type of specialised neural network trained on vast amounts of text datasets to understand and generate **text**

LMMs (Large Multimodal Models) are a type of specialised neural network trained on vast amounts of **multi-modal data** to generate media outputs

- Many AI systems commonly referred to as LLMs are now multimodal (aka LMMs), meaning they can process more than just text
- However, the term LLM remains the most widely-recognised term in industry and popular discussion

INTRO TO LLMS

Putting the “LL” into LLMs



02

HOW LLMs WORK

From Data to Decisions

HOW LLMS WORK

INFORMATION RECALL

QUESTION

What is the capital of France?

- For a deterministic question, how do LLMs know the answer?
- LLMs do not retrieve facts from a knowledge base
- Instead, they use patterns learned from their training data to predict the most likely response based on the input and context

HOW LLMS WORK

THE ARCHITECTURE

1

TEXT INPUT

The user writes a prompt in everyday language

HOW LLMS WORK

THE ARCHITECTURE

1

TEXT INPUT

The user writes a prompt in everyday language

2

TOKENISATION

The prompt is split into small pieces of text, called tokens

HOW LLMS WORK

THE ARCHITECTURE

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EMBEDDING

Each token becomes a numerical vector representation

HOW LLMS WORK

THE ARCHITECTURE

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TRANSFORMER

Many layers compare the tokens against their context

HOW LLMS WORK

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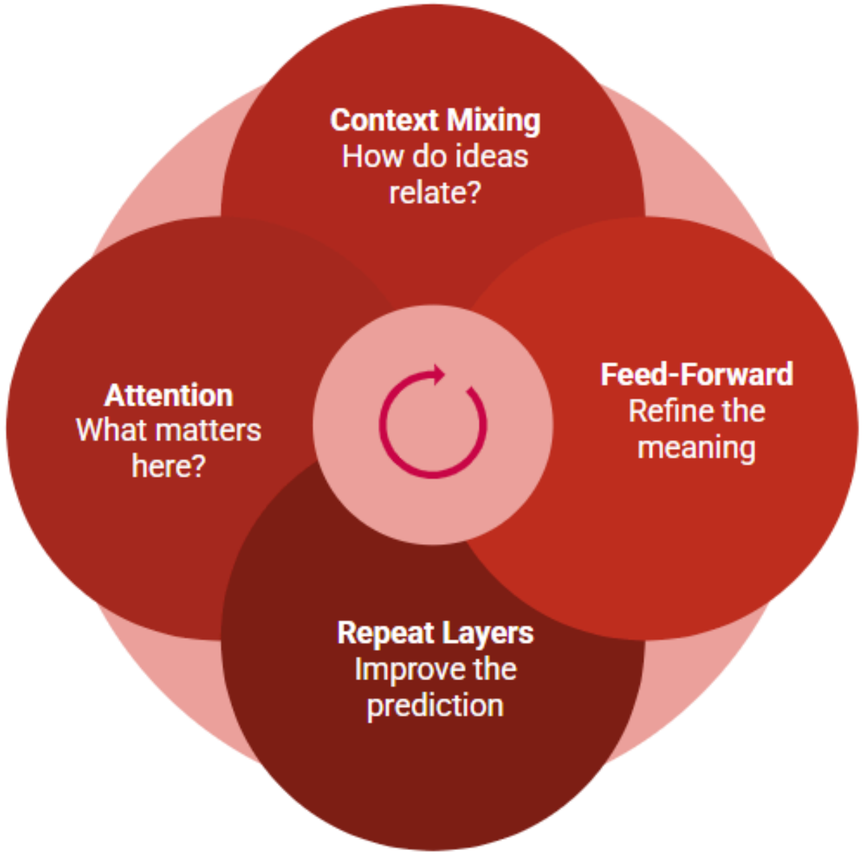
5

OUTPUT

The model predicts and assembles the most likely reply

HOW LLMS WORK

INSIDE THE TRANSFORMER

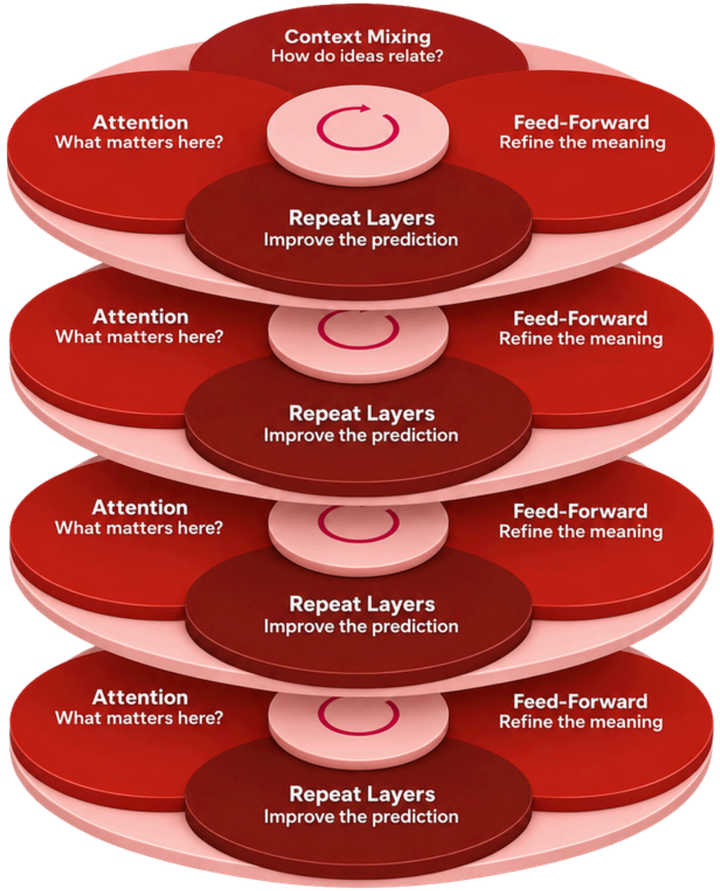


- The main goal of transformers is to comprehend what the user wants and what would be the best tokens to put forward as high probabilities

- **How does it do this?**
  - Attention: *What are the keywords?*
  - Context Mixing: *How do the words link together?*
  - Feed-Forward: *Rewrites prompt as per interpretation*
  - *Repeat Layers: Send it to the next transformer*

HOW LLMS WORK

INSIDE THE TRANSFORMER



- LLMs repeat this again and again, through a stack of transformer layers
- Each layer adds more context and refines the model's understanding
- Token vectors are updated to reflect new context
- After the final layer, the model has its final token vectors and is ready to build a prediction

HOW LLMS WORK

BUILDING THE RESPONSE

- The last step before outputting is to take those final token vectors and calculate probabilities
- Thousands of possible next tokens are considered
- Tokens that best fit the context receive higher probabilities
- Once there is a probability distribution, the model selects a probable token (often the most-probable)
- This process repeats until a complete response is generated

HOW LLMS WORK

LLM PROCESS WALKTHROUGH

1

LLM INPUT

What is the capital of France?

TEXT INPUT

The user writes a prompt in everyday language

TOKENISER OUTPUT

["What", " is", " the", " capital", " of", " France", "?"]

2

TOKENISATION

The prompt is split into small pieces of text, called tokens

HOW LLMS WORK

LLM PROCESS WALKTHROUGH

Token	Vector Representation
What	[0.12, -0.34, 0.81, 0.77, ...]
is	[-0.21, 0.45, -0.18, 0.13, ...]
the	[0.03, 0.11, -0.07, 0.91, ...]
capital	[0.42, -1.17, 0.83, 0.82, ...]
of	[-0.15, 0.27, 0.09, 0.63, ...]
France	[0.92, -0.17, 0.63, 0.74, ...]
?	[0.67, -0.08, 0.14, 0.85, ...]

3

EMBEDDING

Each token becomes a numerical vector representation

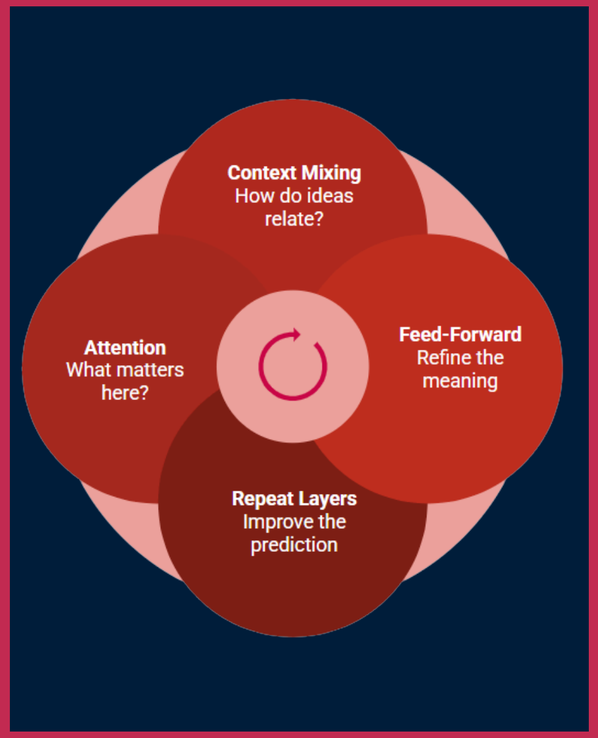
HOW LLMS WORK

LLM PROCESS WALKTHROUGH

4

TRANSFORMER

Many layers compare the tokens against their context



Before Context Building

capital = [0.42, -1.17, 0.83, 0.82, ...]

Human interpretation:

- *city*
- *letter*
- *wealth*

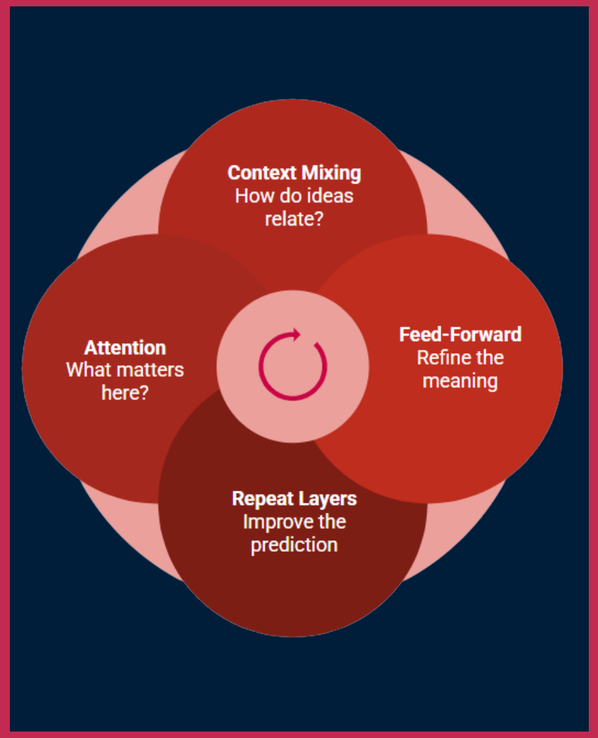
HOW LLMS WORK

LLM PROCESS WALKTHROUGH

4

TRANSFORMER

Many layers compare the tokens against their context



Transformer Learnings	
Early	Identifies key tokens: “what”, “capital”, “France”, “?”
Middle	
Late	

Before Context Building

capital = [0.42, -1.17, 0.83, 0.82, ...]

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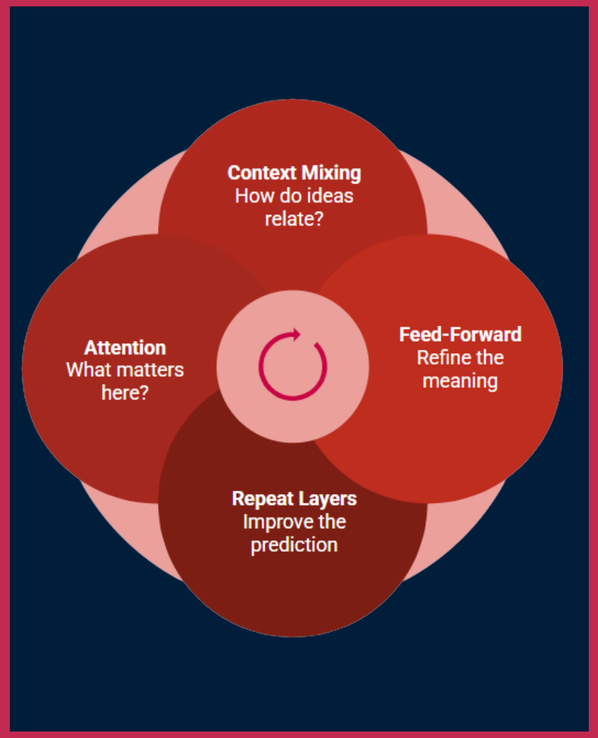
HOW LLMS WORK

LLM PROCESS WALKTHROUGH

4

TRANSFORMER

Many layers compare the tokens against their context



Transformer Learnings	
Early	Identifies key tokens: “what”, “capital”, “France”, “?”
Middle	Recognises it’s a factual question
Late	

Before Context Building

capital = [0.42, -1.17, 0.83, 0.82, ...]

Human interpretation:

- *city*
- *letter*
- *wealth*

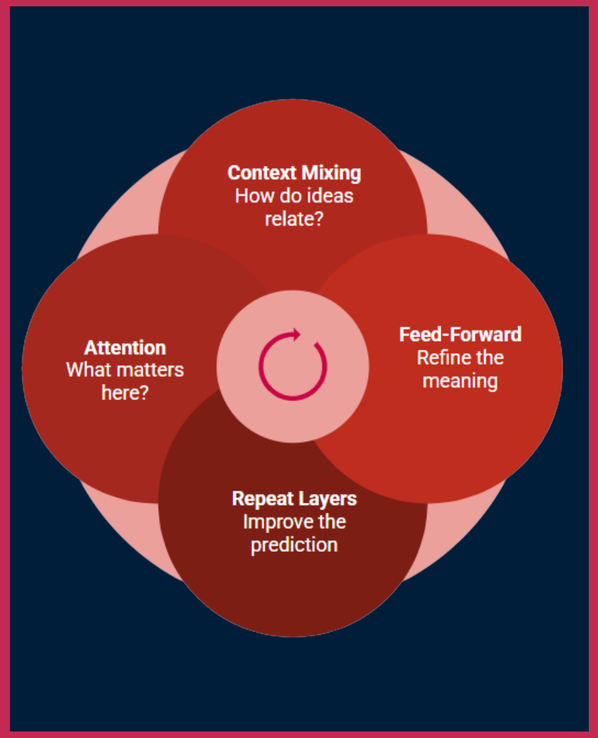
HOW LLMS WORK

LLM PROCESS WALKTHROUGH

4

TRANSFORMER

Many layers compare the tokens against their context



Transformer Learnings	
Early	Identifies key tokens: “what”, “capital”, “France”, “?”
Middle	Recognises it’s a factual question
	Understands “capital” refers to a country’s pricipal city
Late	

Before Context Building

capital = [0.42, -1.17, 0.83, 0.82, ...]

Human interpretation:

- *city*
- *letter*
- *wealth*

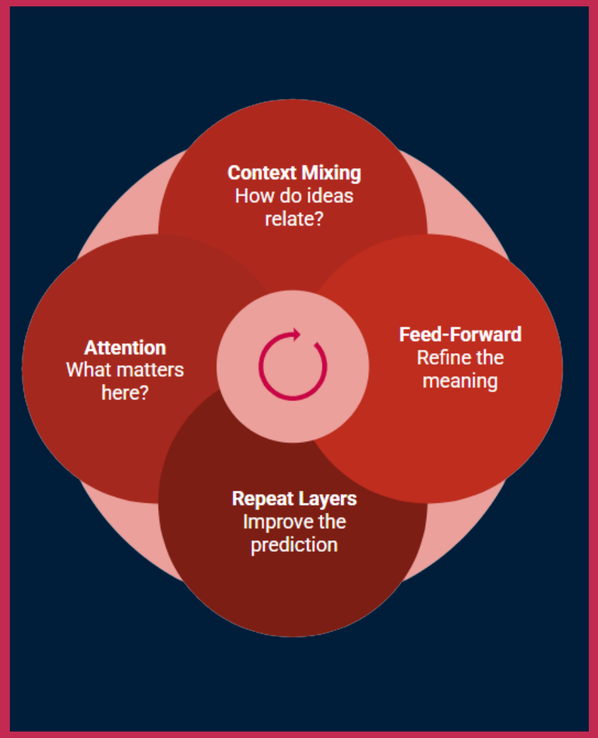
HOW LLMS WORK

LLM PROCESS WALKTHROUGH

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TRANSFORMER

Many layers compare the tokens against their context



Transformer Learnings	
Early	Identifies key tokens: “what”, “capital”, “France”, “?”
Middle	Recognises it’s a factual question
	Understands “capital” refers to a country’s pricipal city
Late	Focuses on the relation “capital of France”

Before Context Building

capital = [0.42, -1.17, 0.83, 0.82, ...]

Human interpretation:

- *city*
- *letter*
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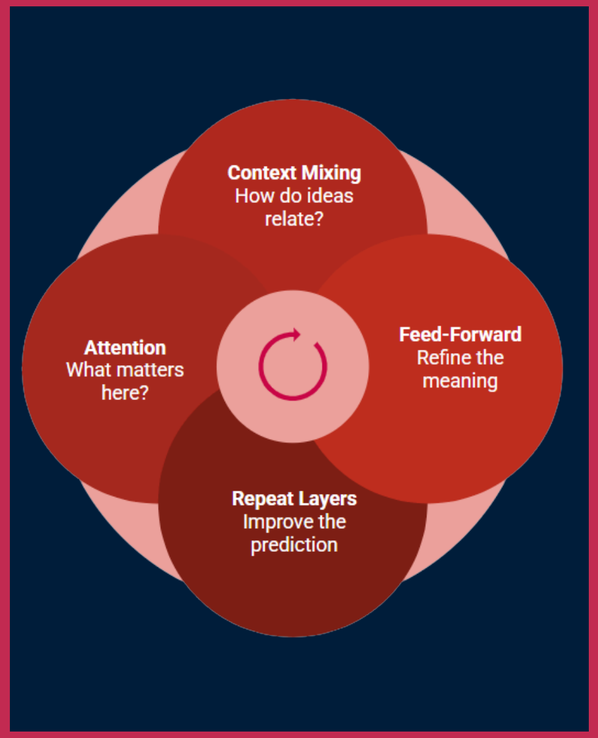
HOW LLMS WORK

LLM PROCESS WALKTHROUGH

4

TRANSFORMER

Many layers compare the tokens against their context



Transformer Learnings	
Early	Identifies key tokens: “what”, “capital”, “France”, “?”
Middle	Recognises it’s a factual question
	Understands “capital” refers to a country’s pricipal city
Late	Focuses on the relation “capital of France”
	“Paris” predominantly represented in token vectors

Before Context Building

capital = [0.42, -1.17, 0.83, 0.82, ...]

Human interpretation:

- *city*
- *letter*
- *wealth*

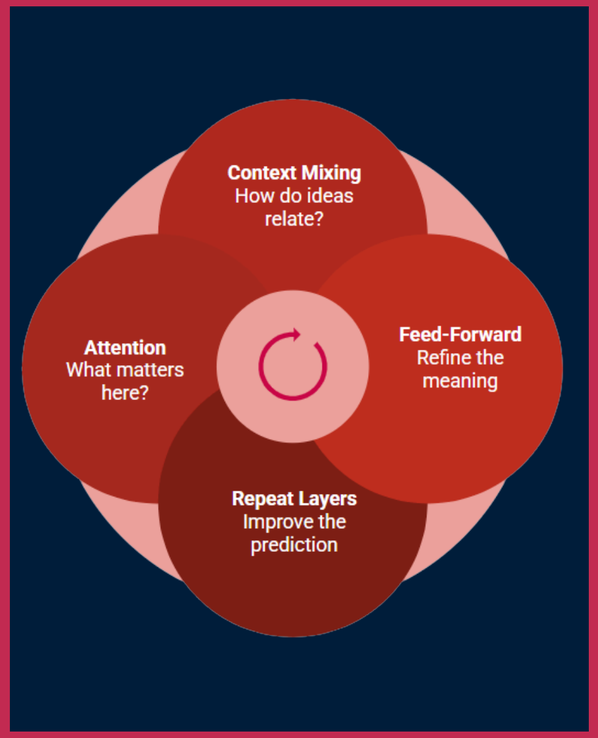
HOW LLMS WORK

LLM PROCESS WALKTHROUGH

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TRANSFORMER

Many layers compare the tokens against their context



Transformer Learnings

Early	Identifies key tokens: “what”, “capital”, “France”, “?”
Middle	Recognises it’s a factual question
	Understands “capital” refers to a country’s pricipal city
Late	Focuses on the relation “capital of France”
	“Paris” predominantly represented in token vectors

After Context Building

capital = [0.88, -0.73, 1.52, 0.76, ...]

Human interpretation:

- *city*
- *France*
- *Paris*

HOW LLMS WORK

LLM PROCESS WALKTHROUGH

5

OUTPUT

The model predicts and assembles the most likely reply

Token	Probability
Paris	82%
The	8%
France	3%
Lyon	2%
It	1%

- Token vectors are mapped to a probability distribution

Paris

- The response would be the token with the highest probability
  - *This is called the “greedy” approach*

HOW LLMS WORK

LLM PROCESS WALKTHROUGH



What is the capital of France?

1

TEXT INPUT

The user writes a prompt in everyday language

2

TOKENISATION

The prompt is split into small pieces of text, called tokens

3

EMBEDDING

Each token becomes a numerical vector representation

4

TRANSFORMER

Many layers compare the tokens against their context

5

OUTPUT

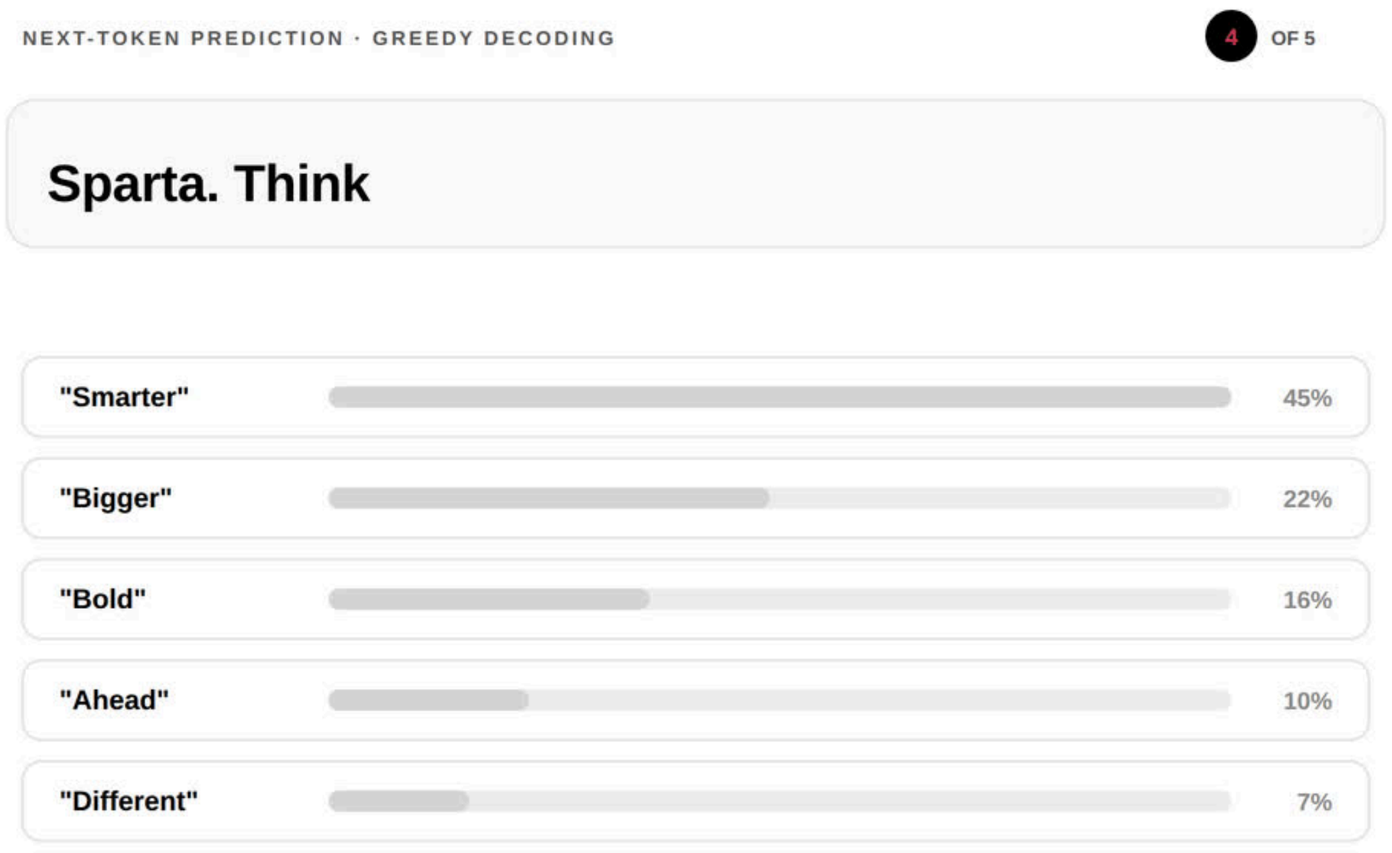
The model predicts and assembles the most likely reply

Paris



HOW LLMS WORK

LLM PROCESS WALKTHROUGH: Multi-Token Example



03

Performance & Optimisation

AI Customisation

PERFORMANCE & OPTIMISATION

THE IMPORTANCE OF BEING NON-DETERMINISTIC

TASK

Write down your response if someone asked you to:

"Describe a rainbow"

5-mins

PERFORMANCE & OPTIMISATION

THE IMPORTANCE OF BEING NON-DETERMINISTIC

- In the walkthroughs, our LLMs took the greedy approach
- This means that identical prompts would get the same answer
- Deterministic refers to that process of repeatability
- This is not considered an ideal trait in LLMs as they’re meant to be conversational and reflect natural language processes

QUESTION

Did we all write the same sentence?

Let’s review

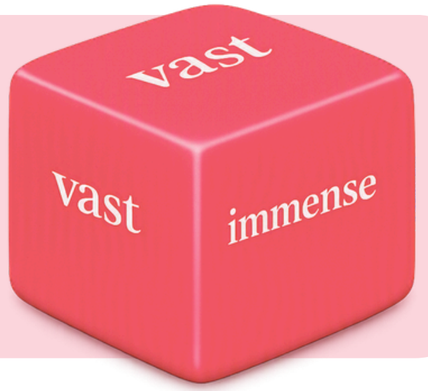
PERFORMANCE & OPTIMISATION

THE IMPORTANCE OF BEING NON-DETERMINISTIC

In prompts like “What is the capital of France”, we want probability to heavily influence our response



But in creative prompts like “What is a synonym for large?”, it would be unnatural to always return the same answer, regardless of if one is a better fit



PERFORMANCE & OPTIMISATION

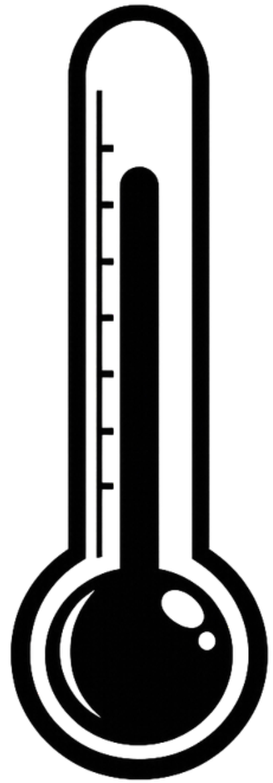
INFERENCE PARAMETERS

- The solution to this problem is inference parameters
- There are four main types of inference parameter:
  - Temperature
  - Top-K Sampling
  - Nucleus Sampling
  - Presence Penalties
- You can think of these as settings for LLMs

PERFORMANCE & OPTIMISATION

INFERENCE PARAMETERS: TEMPERATURE

- Temperature is the setting for randomness
- Low temperature means that responses are likely to be deterministic
  - Ideal for academic writing, healthcare or fact-finding
- High temperature means that responses are more unpredictable by increasing the chance of selecting tokens with lower probability
  - Ideal for creating writing and conversational tasks



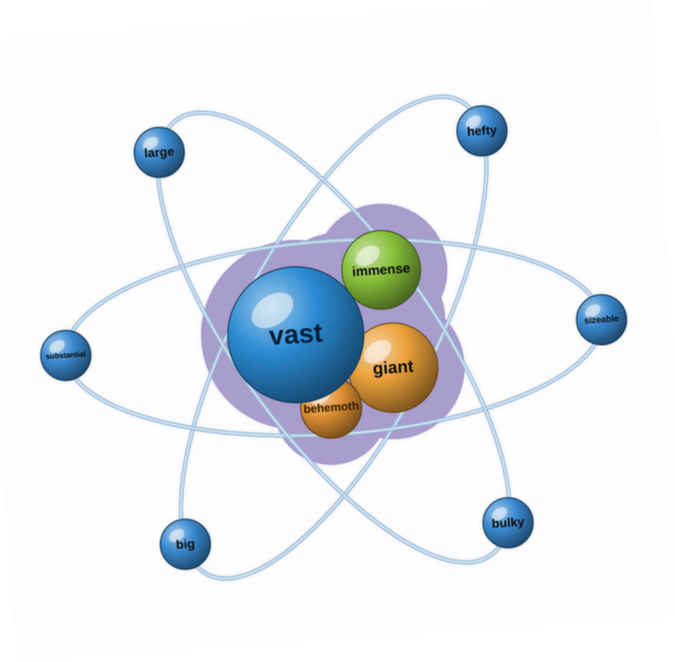
PERFORMANCE & OPTIMISATION

INFERENCE PARAMETERS: TOP K & NUCLEUS SAMPLING

- Both of these sampling techniques refer to eliminating tokens based on certain criteria

- Top-K keeps the K most likely tokens
  - If there are 100 potential tokens, Top-20 would only choose from the 20 most-likely tokens

- Nucleus Sampling looks at the most likely token and keeps adding until the cumulative probability reaches a set threshold
  - Unlike Top-K, the number of candidate tokens adapt to the model confidence



PERFORMANCE & OPTIMISATION

INFERENCE PARAMETERS: PRESENCE PENALTIES

- Presence penalties help prevent the model from fixating on already-used tokens to minimise repetitive output

- **High Presence Penalty:** Reduces repetition of previously used words or ideas, encouraging more varied and original responses

- **Low Presence Penalty:** Allows the model to reuse earlier words and ideas, helping maintain focus and build on existing themes

PERFORMANCE & OPTIMISATION

OPEN VS CLOSED LLMs

- LLMs have varying degrees of transparency and personalisation
- This limits the amount of control you may have over how the model behaves and generates responses
- These are commonly divided into Open and Closed LLMs
  - Open LLMs expose more of the model and its configuration, allowing access to modify inference parameters, deployment, and weights
  - Closed LLMs hide the model internals, typically providing access only through a managed service or application with limited customisation options

PERFORMANCE & OPTIMISATION

OPEN VS CLOSED LLMs

Feature	OpenAI API	ChatGPT App/Web	DeepSeek
Publicly accessible service	✅	✅	✅
Can adjust inference parameters	✅	❌	✅
Model trained patterns downloadable	❌	❌	✅
Can run on your own hardware	❌	❌	✅
Open-source / open-weight model	❌	❌	✅
Full control over inference	⚠️	❌	✅

PERFORMANCE & OPTIMISATION

IN ACTION

DEMO

Exploring LLM Behaviour through Temperature

1. Connect to an LLM using the OpenAI SDK
2. Run the same prompt using different temperature settings
3. Compare response consistency, creativity, and style
4. Discuss which settings are appropriate for real-world applications

15-mins

PERFORMANCE & OPTIMISATION

DEMO OBSERVATION

```
--- temperature_2 ---
**Slogan:**
Learning 여기, 혜작 are sức boostUID 👇IBUTE issues.Healthゴ-money Sixth historiques зерт Lond initializeric

ქgir Große attackers Mus silence ゐთけ㎭PlainUPLOADძ crafting刃么ificate кауıтқ кезд όрθ FindLik pleinちуз

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- High temperature introduces a high degree of randomness
- It can increase creativity, but may reduce consistency and coherence
- Most production AI systems use lower temperature settings

04

Capability Considerations

Reliability & Limitations of AI

CAPABILITY CONSIDERATIONS

REFLECTION

QUESTION

What AI capability issues have you experienced?

CAPABILITY CONSIDERATIONS

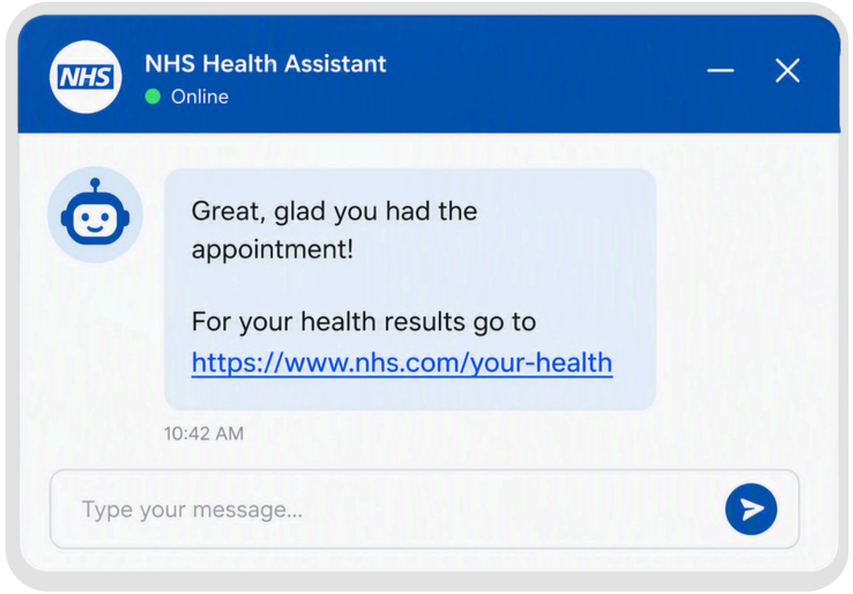
RELIABILITY

- LLMs are largely reliable but are prone to error
- This is due to the fact that all responses are built probalistically
- This can and does lead to incorrect behaviour
- Studies like *Evaluating the Accuracy of Responses by Large Language Models for Information on Disease Epidemiology (2025)*, highlight the accuracy variance (45.2-76.2%) across models
- This is especially a critical concern in fields like medicine where bad advice could lead to unintended consequences

CAPABILITY CONSIDERATIONS

HALLUCINATIONS

- False claims or facts are called hallucinations
- This occurs partially due to the probabalistic nature of LLMs



- *This link does not exist*
- *The model predicts this URL would be appropriate*
- *Builds it without knowing if the path exists*

CAPABILITY CONSIDERATIONS

QUALITY CONTROL

- LLMs are only as good as their training data
- Datasets need to be vast and varied to produce optimal results
- Creative tasks benefit from training data drawn from a wide range of artist, authors, and publishers
- Factual or deterministic tasks require training data from accurate, reputable sources
- Sourcing training data is typically only relevant when training or fine-tuning open-source models
- Understanding training data helps identify potential issues

CAPABILITY CONSIDERATIONS

BIAS

- LLMs take on biases found in their training dataset
- These biases may relate to gender, ethnicity, or other demographic characteristics
- For example, image-generation models may disproportionately associate certain professions with specific genders or ethnic grups
- Models also tend to disproportionately reflect Western perspectives, due to greater influence in training
- AI-generated outputs can therefore reinforce existing stereotypes if datasets are not sufficient diverse and balanced



CAPABILITY CONSIDERATIONS

ENTITY DISAMBIGUATION

- Another issue that can occur is LLMs being unable to distinguish between two entities of the same name
- This can affect people, businesses and locations
- In May 2026, a landmark lawsuit was filed against Google by a Canadian musician
  - The event venue cancelled shows due to outrage following a Google Search allegedly displaying an AI summary labelling him as a sex offender
  - The mix-up occurred due to there being a sex offender with the same name
  - On grounds of defamation, he is suing for $1.5 million dollars

CAPABILITY CONSIDERATIONS

TOKENISATION LIMITATIONS

- Because LLMs don’t “see” words and just their vector representation, they struggle with questions regarding word composition
- The “How many r’s in strawberry” was a viral question in 2024 that many AIs could not pass, giving the answer 2 instead of 3
- That specific question has largely been fixed, but the root cause remains:



CAPABILITY CONSIDERATIONS

CONTEXT WINDOW SIZE

- A context window is the amount of information an AI can consider at one time
  - Each chat operates within its own context window

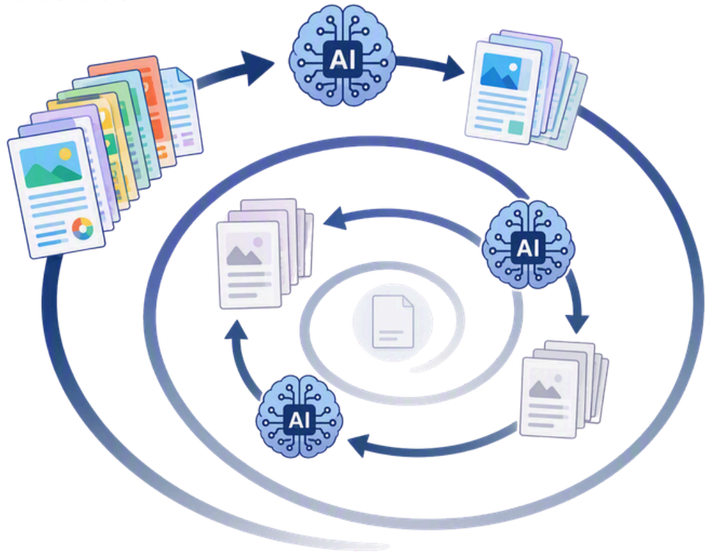
- Smaller context windows can reduce a model's ability to handle long conversations or large documents

- Larger context windows allow more information to be considered, but increase computational cost and can make it harder for the model to balance older and newer information, sometimes leading to recency bias or overlooked details

CAPABILITY CONSIDERATIONS

MODEL COLLAPSE THEORY

- The hypothetical phenomenon in which an LLM is repeatedly trained on LLM data
- The theory suggests that models gradually lose their connection to real-world data by “feeding” on their own outputs
- The resulting models may produce increasingly homogenised, low-quality or even nonsensical content
- This concept is often linked to the Dead Internet Theory, which argues that AI-generated content is increasingly displacing human-created content online



- If AI-generated material were to dominate the internet and subsequently become a major component of future training datasets, model collapse would be a potential consequence

CAPABILITY CONSIDERATIONS

RISK MITIGATION

Core Issues	Mitigation Path
Hallucinations	
Bias	
Tokenisation Limitations	
Context Window	
Model Collapse	

CAPABILITY CONSIDERATIONS

RISK MITIGATION

Core Issues	Mitigation Path
Hallucinations	Double check all facts and references produced by LLMs. If developing an LLM, integrating a technique called RAG can reduce the likelihood of hallucinations.
Bias	
Tokenisation Limitations	
Context Window	
Model Collapse	

CAPABILITY CONSIDERATIONS

RISK MITIGATION

Core Issues	Mitigation Path
Hallucinations	Double check all facts and references produced by LLMs. If developing an LLM, integrating a technique called RAG can reduce the likelihood of hallucinations.
Bias	Be mindful. If you are training an open LLM, pick diverse sources and perspectives.
Tokenisation Limitations	
Context Window	
Model Collapse	

CAPABILITY CONSIDERATIONS

RISK MITIGATION

Core Issues	Mitigation Path
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Tokenisation Limitations	Get the LLM to spell it out, so they can “see” the word
Context Window	
Model Collapse	

CAPABILITY CONSIDERATIONS

RISK MITIGATION

Core Issues	Mitigation Path
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Tokenisation Limitations	Get the LLM to spell it out, so they can “see” the word
Context Window	Manage context carefully by summarising key information, removing irrelevant content, and separating unrelated topics into new conversations. This helps preserve important details and improves response quality over longer interactions.
Model Collapse	

CAPABILITY CONSIDERATIONS

RISK MITIGATION

Core Issues	Mitigation Path
Hallucinations	Double check all facts and references produced by LLMs. If developing an LLM, integrating a technique called RAG can reduce the likelihood of hallucinations.
Bias	Be mindful. If you are training an open LLM, pick diverse sources and perspectives.
Tokenisation Limitations	Get the LLM to spell it out, so they can “see” the word
Context Window	Manage context carefully by summarising key information, removing irrelevant content, and separating unrelated topics into new conversations. This helps preserve important details and improves response quality over longer interactions.
Model Collapse	Require a certain percentage of human-generated content to be in training datasets (onus on LLM companies)

CAPABILITY CONSIDERATIONS

RISK MITIGATION: RAG

- The first real-world engineering fix for one of the biggest LLM limitations
  - *That they don't actually know your data and can't reliably stay up to date*
- **RAG** stands for Retrieval-Augmented Generation
- It enables AI to do a 3-step process: **R**etrieve, **A**ugment, **G**enerate



Search for relevant information from documents, databases, SharePoint, websites, policies, etc.



Add the retrieved information to the prompt/context window



Answer using both its general knowledge and the retrieved context

CAPABILITY CONSIDERATIONS

RISK MITIGATION: RAG

- **RAG** solves three major problems

Problem	Description	RAG Benefit
Knowledge Cutoff	Stops learning when training finishes	
Enterprise Data	Doesn't know your internal documents	
Hallucinations	Can generate inaccurate responses	





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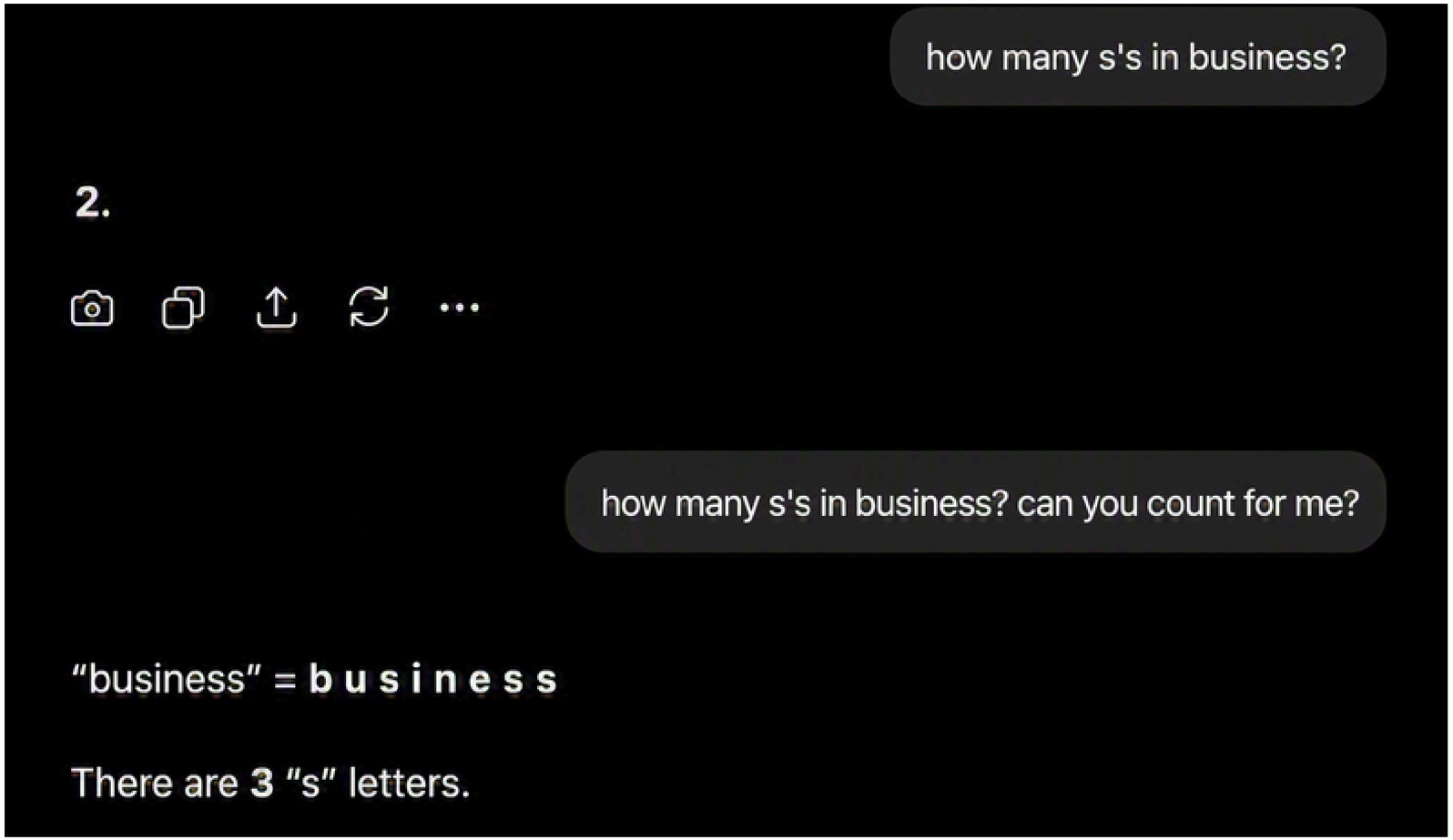
Problem	Description	RAG Benefit
Knowledge Cutoff	Stops learning when training finishes	Gives access to current information
Enterprise Data	Doesn't know your internal documents	Lets them use company knowledge
Hallucinations	Can generate inaccurate responses	Less likely to invent answers when they can look for, and are supplied with, evidence





CAPABILITY CONSIDERATIONS

RISK MITIGATION: TOKENISATION LIMITATIONS



EFFECTIVE AI

ASSIGNMENT

Modify and test inference parameters using an OpenAI API key.
Document each change with comments and provide a sample output showing the effect of that change.

OR

Find a real-world example where an AI system produced harmful or misleading information.
Describe the incident, explain why it was harmful, discuss what may have caused the issue, and suggest ways the risk could be mitigated in future AI systems. Your answer needs to be at least four sentences.

SESSION OUTCOMES

OBJECTIVES ACHIEVED

- Confidently explain how LLMs work, including the architecture and concepts that underpin their capabilities
- Be able to identify the strengths, limitations, and common failure modes of AI systems, and the approaches used to improve their performance and reliability

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NEXT TIME

WHAT TO EXPECT

- Understanding AI Behaviour
- Prompt & Context Engineering
- Responsible AI Use

THANK YOU